INTERIM FEDERAL SPECIFICATION

TAPE, RECORDING, AUDIO, 1/4-INCH MAGNETIC OXIDE-COATED

This Interim Federal Specification was developed by Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the detailed requirements for 1/4-inch magnetic recording tape, wound on reels or hubs, designed for use in audio recording applications.

1.2 Classification. Tape types covered by this specification are designated in the following form, as specified (see 6.1):

TAT	AB	HR	250	NA	I	1200
Basic indicator (see 1.2.1)	Oxide characteristic indicator (see 1.2.2)	Backing and thickness indicator (see 1.2.3)	Width indicator (see 1.2.4)	Reel and Hub indicator (see 1.2.5)	Reel and Hub Type indicator (see 1.2.6)	Length indicator (see 1.2.7)

1.2.1 Basic indicator. The basic indicator defines the general application for tape, as specified herein. The basic indicator is "TAT" for tape, audio type.

1.2.2 Oxide characteristic indicator. The oxide characteristic indicator, which consists of a letter or series of letters, identifies a particular oxide by its physical or magnetic characteristics, as specified herein.

1.2.3 Backing and thickness indicator. The backing and thickness indicator combines two letters as an indicator. The first letter identifies the type of backing material, and the second letter indicates the nominal thickness of the backing. The backing material is indicated by the letter "H" for polyester base film and "F" for cellulose acetate base film. The backing material of a nominal thickness of 1.5 mils (thousandths of an inch) is defined by the letter "R" for "regular", and the backing material of a nominal thickness of 1.0 mils is defined by the letter "T" for "thin."

1.2.4 Width indicator. The width indicator is the width of the tape in thousandths of an inch (1/4-inch tape is 250), (see 3.2.2).

1.2.5 Reel and hub indicator. The reel and hub indicator is identified by letters in accordance with 3.1.4.4.

1.2.6 Reel and hub type indicator. The reel and hub type indicator is identified by letters in accordance with 3.1.4.4.

1.2.7 Length indicator. The length indicator is the nominal length, in feet, of a tape wound on a reel or hub (see 3.2.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-378 - Plastic Sheet and Strip (Polyolefin).
W-R-175 - Reels and Hubs for Magnetic Recording Tape, General Specification for.
W-R-175/1 - Reels, Standard, Plastic, and Fiberglass, 5/16-inch Center Hole.
W-R-175/2 - Reel Hubs, Standard, Fiberglass and Metallic, 3-inch Center Hole.
W-R-175/3 - Reels, Standard, Fiberglass and Metallic, 3-inch Center Hole.

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Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 406 - Plastics: Methods of Testing.
Fed. Test Method Std. No. 406/1013 - Tensile Properties of Thin Sheets and Films.
Fed. Test Method Std. No. 406/4041 - Electrical Resistance (Surface).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from the established distribution points in their agencies.)

Military Specifications:

MIL-I-631 - Insulation, Electrical, Synthetic - Resin Composition, Nonrigid.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories and Repair Parts, Packaging and Packing of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids, or request for proposal, shall apply:

American Society for Testing and Materials (ASTM) Standards:

D-638 - Tensile Properties of Plastics: Method of Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 General requirements.

3.1.1 Product qualification. All audio magnetic tape supplied under this specification shall be products which have been tested and approved for listing on the applicable Qualified Products List (QPL) maintained by the General Services Administration (GSA-FSS). Hereinafter the term qualifying activity shall mean the GSA.

3.1.1.1 Qualified products listing. The Qualified Products List shall consist of products which have been tested and passed those qualification tests specified herein.

3.1.1.2 Qualification product testing. Products for test may be submitted by prospective suppliers of magnetic tape on dates to be announced by the General Services Administration. Should the submitted product(s) fail to meet the specification as defined herein, the qualifying activity reserves the right to refuse to accept such products for additional qualification tests until satisfactory data and test results have been submitted indicating correction of product deficiencies. The qualifying activity reserves the right to levy a charge to cover cost of testing for qualification of products.

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3.1.1.3 Qualification withdrawal. A supplier's qualified product listing may be withdrawn in accordance with "Provisions Governing Application by Manufacturers for Inclusion on Federal Qualified Products Lists (QPL's)" (see 6.3).

3.1.1.4 Requalification. A supplier's product, once withdrawn from the Qualified Products List, shall not be accepted for requalification until satisfactory data and test results have been submitted to the qualifying activity by the supplier indicating correction of the product fault(s). The qualifying activity reserves the right to levy a charge to cover cost of testing for requalification of products.

3.1.1.5 Process change. Qualification of a supplier's product under this specification, once established, applies only to those tapes manufactured by the specific process in use at the time of qualification. All proposed process changes shall be reported to the qualifying activity with a statement by the supplier as to the general nature of the changes and the extent and effect of such changes on the delivered product. The qualifying activity reserves the right to require six weeks notice, and sufficient samples of the new product for qualification tests, from the supplier prior to delivery of products manufactured under any such change in process. The foregoing requirement does not in any way relieve the manufacturer of delivery requirements, quality control, and testing necessary to insure that all products delivered under this specification are equal to those products originally submitted for qualification.

3.1.2 Minimum lot size. The minimum lot size **shall** be 500 reels of tape for orders of 500 reels of tape or more. For orders of less than 500 reels of tape, the lot size shall be the order size. Orders may be grouped into larger lots or may be shipped from previously approved standby stock.

3.1.2.1 Definition of lot. A lot consists of all tape, selected for delivery by the supplier, having been manufactured and processed from the same batch or mix of the basic coating materials used in the production of the magnetic tape. This batch or mix shall be sufficient to insure compliance with the requirements of 3.1.2. In the case of a continuous batching or production process, a lot shall consist of sufficient tapes selected from concurrent production to meet the requirements of 3.1.2.

3.1.3 Component parts. Tape delivered under this specification shall be supplied on reels or hubs that meet the requirements of 3.1.4.4 and that are packaged in accordance with 5.1, as specified (see 6.1).

3.1.4 Materials.

3.1.4.1 Tape. The tape supplied under this specification shall be new and shall consist of a layer of magnetic material on a suitable backing material and shall have the physical, magnetic and other characteristics, as specified herein.

3.1.4.2 Toxic compounds. The tape shall not produce toxic effects under conditions normally encountered in Government service.

3.1.4.3 Inflammable materials. Tape which will ignite from a match flame, and when so ignited continue to burn in a still carbon dioxide atmosphere, shall not be supplied under this specification.

3.1.4.4 Reels and hubs. Reels and hubs shall conform to the requirements of W-R-175, unless otherwise specified (see 6.1).

3.1.5 Wind. The tape shall be wound on the reel or hub with the oxide surface facing toward the hub.

3.2 Dimensional requirements.

3.2.1 Length. The tape shall be supplied in minimum lengths in accordance with table I, unless otherwise specified (see 6.1), on reels or hubs that conform to the requirements of 3.1.4.4.

3.2.2 Width. The tape width shall be 0.246 inches ± 0.002 inches.

3.2.3 "E" value. The minimum "E" value for all tape lengths shall be in accordance with table II. The "E" value is defined as the radial distance by which the reel flanges extend beyond the outer-most layer of tape when the tape is wound at a tension of from 3 to 5 ounces on a reel that meets the requirements of 3.1.4.4.

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TABLE I. Tape length

Reel diameter (inches)	Hub diameter (inches)	1.5 Mil base		1.0 Mil base	
		Length indicator	Minimum length (feet)	Length indicator	Minimum length (feet)
3	---	150	155	----	----
4	---	300	305	----	----
5	---	600	610	900	915
7	---	1200	1215	1800	1820
10.5	4.5	2500	2520	3600	3620
14	4.5	5000	5025	7200	7230

TABLE II. Minimum "E" value

Length indicator	Minimum "E" value (inches)
150 to 1000	0.10
1000 to 5000	0.20

3.3 Physical requirements.

3.3.1 Yield strength. The 1 percent offset yield point, the force at 3 percent elongation, and the breaking force, of a particular type of tape shall equal or exceed the value specified in table III when tested in accordance with 4.4.4.

3.3.2 Elongation under stress. The elongation of a particular type of tape shall not exceed the value specified in table III when tested in accordance with 4.4.5.

3.3.3 Humidity stability. The humidity stability (cupping) of a particular type of tape shall not exceed the value specified in table III when tested in accordance with 4.4.6. Cupping is defined as the transverse curvature of a strip of tape viewed "end on."

3.3.4 Fungus resistance. Tape shall not support fungus growth. The tape shall be considered fungus-resistant if at least 2 of 3 specimens are rated "0" or "1" when tested as specified in 4.4.7.

3.3.5 Layer-to-layer adhesion. The tape shall show no sticking or layer-to-layer adhesion when tested in accordance with 4.4.8. (Layer-to-layer adhesion, commonly termed blocking, is defined as that property of a magnetic tape wherein one layer when held in close proximity to an adjacent layer, as when in a reel of tape, exhibits an adhesive nature and bonds itself to an adjacent layer so that free and smooth separation of the layers is difficult. In practice, the amount of adhesion may vary from a quantity which merely prevents adjacent layers from separating freely to the most severe condition in which the adhesion is so great that the magnetic coating is disengaged from the tape backing when the tape is unwound. Adhesion in any form is objectionable, since tape which displays even mild layer-to-layer adhesion may in time progress to the point where it is impossible to separate layers without destroying the tape. In operation, adhesion can result in excessive flutter, timing errors and poor uniformity of output level.)

3.3.6 Longitudinal curvature. The deviation of each tape edge from a straight line shall not exceed 0.125 inches when measured in accordance with 4.4.9.

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TABLE III. Physical requirements

Characteristic	Backing and thickness indicator (see 1.2.3)			Units (1/4-inch of tape width)
	HR	HT	FR	
Yield strength	4.9	3.2	4.0	Pounds
Elongation under stress	0.30	0.50	1.0	Percent
Humidity stability	5	5	20	Degrees

3.3.7 Wound pile envelope. The tape as initially received from the manufacturer shall be smoothly wound to form an integral mass and shall not be loose either axially or circumferentially. There shall be no visible folds, buckling, cinching, spoking, or gaps between the tape layers. The edges of the wound tape shall be in a single plane with a minimum of protruding tape edges, layers, or groups of layers. The tape shall be wound in such a manner that it will not be physically damaged on either edge resulting in functional failure (see 4.4.10).

3.3.8 Anchorage. The magnetic layer of the tape shall show no visible evidence of anchorage failure after completion of all tests (see 4.4.11). Anchorage failure may be manifested as a separation of the magnetic coating from the base material of the tape, or internal failure of the coating layer itself. The tape shall not be affected by commonly used head cleaners.

3.3.9 Splices. All tape lengths supplied under this specification shall be continuous and splice free (see 4.4.12).

3.3.10 Workmanship and general examination. The magnetic tape, reels, and containers shall be manufactured and processed in a careful and workmanlike manner in accordance with good practice. Both the magnetic oxide surface and base shall be free from any raised areas, dust, flakes, powder, holes, foreign matter, scratches, creases and other defects which would render the tape unsuitable for its intended use. The condition of tape slitting, reels, wind, and containers shall be examined for workmanship and cleanliness (see 4.4.13).

3.4 Magnetic requirements.

3.4.1 Sensitivity. The output of a particular type of tape at any point in a reel, when measured in accordance with 4.5.4.1, shall not vary from the standard output level by more than the value specified in table IV.

3.4.2 Frequency response. The output at the specific test frequencies of a particular type of tape, when measured in accordance with 4.5.4.2, and when normalized to the 1,000 Hz output and compared with the response of the primary reference tape, shall fall within the limits specified in table IV.

3.4.3 Harmonic distortion. The third harmonic distortion at the specific test record levels of a particular type of tape, when measured in accordance with 4.5.4.1 and 4.5.4.4, shall not exceed the value specified in table IV.

3.4.4 Output level uniformity. The uniformity of the signal output at the specific test frequencies of a particular type of tape, when tested as specified in 4.5.4.3, shall be such that the maximum deviation from the maximum peak output does not exceed the value specified in table IV.

3.4.5 Signal to direct current (dc) noise ratio. The signal to dc noise ratio of a particular type of tape, when tested in accordance with 4.5.4.4, shall equal or exceed the value specified in table IV.

3.4.6 Layer-to-layer-signal transfer. A signal resulting from layer-to-layer signal transfer for a particular type of tape, when measured as specified in 4.5.4.5, shall be below the signal level by at least the value specified in table IV.

3.4.7 Ease of erasure. An erase field of the specified test value (see 4.5.4.6), shall effect a reduction in signal for a particular type of tape of at least the value specified in table IV, when tested as specified in 4.5.4.6.

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3.4.8 Frictional vibration. The tape shall not produce squeal components of an audible nature during the magnetic tests. Frictional vibration, which in practice manifests itself as a squeal, is defined as the random frequency and amplitude modulation, or both, of a recorded signal which results when the tape passes over the guides and heads of a magnetic tape recorder. This results from the changing frictional components introduced between the tape, heads, and guides as the tape travels past them.

TABLE IV. Magnetic requirements

Characteristic	Oxide Characteristic Indicator AB	Units
Sensitivity	0 ± 2	Decibels
Frequency response at:		
200 Hz	0 ± 2	Decibels
5,000 Hz	0 ± 2	Decibels
10,000 Hz	0 - 2, (No + limit)	Decibels
Harmonic distortion at:		
Standard record level	1.0	Percent
Maximum record level	4.0	Percent
Output level uniformity at:		
1,000 Hz	25	Percent
8,000 Hz	30	Percent
Signal to dc noise ratio	52	Decibels
Layer-to-layer signal transfer for:		
HT	40	Decibels
HR	46	Decibels
FR	46	Decibels
Ease of erasure	60	Decibels

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) Qualification testing (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).
(c) Acceptance inspection (see 4.2.3).
(d) Examination of preparation for delivery (see 4.2.4).

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4.2.1 Qualification testing. Qualification testing shall be conducted at a laboratory designated by the General Services Administration, Federal Supply Service, and shall consist of tests listed in 4.3, 4.4, and 4.5. All samples subjected to these tests must conform to the associated requirements listed in section 3 of this specification.

4.2.1.1 Qualification of a particular tape base, thickness, and oxide characteristic shall constitute qualification of all tapes of that particular base, thickness, and oxide characteristic regardless of length.

4.2.1.2 Samples for qualification. The sample size shall consist of ten (10) reels of 1/4-inch 1200-foot tape lengths for each particular tape base, thickness, and oxide characteristic for which qualification is desired. All tape samples shall be submitted on reels which conform to the requirements of 3.1.4.4. The manufacturer shall also submit test data showing that each tape type for which qualification is desired has successfully met all the requirements of this specification.

4.2.2 Quality conformance inspection. In accordance with 4.1, the contractor shall be responsible for the performance of the quality conformance inspections. The contractor shall select a sample number of reels of tape from each lot of tape offered for delivery. Sample reels of tape shall be selected in accordance with 4.2.2.2 and the contractor shall subject the samples to the appropriate examinations and tests of table V to determine compliance with the specification requirements. All necessary test equipment and facilities required to conduct the subject examinations and tests shall be furnished by the contractor. Alternate test techniques and equipment may be used by the contractor, subject to review and approval by the qualifying activity that such techniques and equipment are in fact adequate to insure compliance with specification requirements.

4.2.2.1 Reporting of quality conformance test results. Unless otherwise directed by the contracting officer (see 6.1), the contractor shall maintain a complete record of all production test results for the duration of the contract. The production test records shall be available to the Government at all reasonable times. The records shall include the information necessary to identify the lot, the testing equipment, the inspector, and the date of test.

4.2.2.2 Sampling for quality conformance inspection. Sampling procedures for quality conformance inspection shall be in accordance with MIL-STD-105 and as specified herein. The sample reels of tape shall be subjected to the Group A, B, C, and D tests shown in table V.

4.2.2.2.1 Sampling for group A examinations and tests. The sampling procedure for group A examinations and tests shall be in accordance with MIL-STD-105, special inspection level S-2 at an acceptance quality level (AQL) of 6.5 percent defective.

4.2.2.2.2 Sampling for group B examinations and tests. The sampling procedure for group B examinations and tests shall be in accordance with MIL-STD-105, special inspection level S-2 at an acceptance quality level of 6.5 percent defective.

4.2.2.2.3 Sampling for group C examinations and tests. The sampling procedure for group C examinations and tests shall be in accordance with MIL-STD-105, special inspection level S-2 at an acceptance quality level of 4.0 percent defective.

4.2.2.2.4 Sampling for group D examinations and tests. Group D examinations and tests will be required initially and as often as deemed necessary to assure continued compliance with the specified requirements. Ten sample reels from a lot shall be subjected to the group D examinations and tests shown in table V. Failure to conform to any requirement of the group D examinations and tests shall be cause for the rejection of that lot represented by the samples.

4.2.3 Acceptance inspection. Acceptance testing on shipment lots shall be conducted by the qualifying activity. Acceptance tests shall consist of those group A and B examinations and tests specified in table V.

4.2.3.1 Sampling for acceptance inspection. The sampling procedures for the group A and group B examinations and tests shall be as specified in 4.2.2.2.1 and 4.2.2.2.2.

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TABLE V. Inspection tests

Test Group	Examinations and Tests	Requirement Paragraph	Test Paragraph
A	Length	3.2.1	4.3.1
	Width	3.2.2	4.3.2
	E Value	3.2.3	4.3.3
	Wound pile envelope	3.3.7	4.4.10
	Anchorage	3.3.8	4.4.11
	Splices	3.3.9	4.4.12
	Workmanship and general examination	3.3.10	4.4.13
	Output level uniformity	3.4.4	4.5.4.3
B	Sensitivity	3.4.1	4.5.4.1
	Frequency response	3.4.2	4.5.4.2
	Signal to dc noise ratio	3.4.5	4.5.4.4
	Frictional vibration	3.4.8	-------
C	Yield strength	3.3.1	4.4.4
	Layer-to-layer adhesion	3.3.5	4.4.8
D	Toxic compounds	3.1.4.2	-------
	Inflammable materials	3.1.4.3	-------
	Reels and hubs	3.1.4.4	-------
	Wind	3.1.5	-------
	Elongation under stress	3.3.2	4.4.5
	Humidity stability	3.3.3	4.4.6
	Fungus resistance	3.3.4	4.4.7
	Longitudinal curvature	3.3.6	4.4.9
	Harmonic distortion	3.4.3	4.5.4.1 and 4.5.4.4
	Layer-to-layer signal transfer	3.4.6	4.5.4.5
	Ease of erasure	3.4.7	4.5.4.6

4.2.4 Examination of preparation for delivery. Interior and exterior examinations shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5, as specified (see 6.1). A sample unit for these examinations shall be one shipping container, fully prepared for delivery. The sample units shall be selected from end item inspection lots and examined for the defects listed in table VI. The sampling procedure shall be in accordance with MIL-STD-105, Special Inspection Level S-2 at an acceptance quality level of 2.5 percent defective. The lot size shall be taken as the number of shipping containers in the end item inspection lot.

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TABLE VI. Preparation for delivery defects

Examine	Defects
Interior packaging.	Use of improper or defective materials. Quantity in unit container not as specified. Quantity in shipping container not as specified. Incorrect packaging method employed. Cushioning or padding omitted. Cushioning inadequate for the physical protection of the item. Unsealed, punctured, or improperly sealed bag, wrap, or envelope. Damaged or otherwise defective reels.
Exterior packing and marking.	Use of improper or defective materials. Type, grade, class and style of shipping container not as specified. Marking incorrect; incomplete; illegible; omitted; of improper size, location, sequence, or method of application. Inadequate application of components such as: incomplete closure of case liners and container flaps; loose or inadequate sealing, strapping, or stapling. Bulged or distorted container.

4.3 Dimensional measurements.

4.3.1 Length. The tape length shall be measured to determine conformance to the requirements of 3.2.1.

4.3.2 Width. The tape width shall be measured to determine conformance to the requirements of 3.2.2.

4.3.3 "E" value. The "E" value shall be measured to determine conformance to the requirements of 3.2.3.

4.4 Physical tests and measurements.

4.4.1 Preliminary conditioning. Preliminary conditioning of each reel or hub of tape will be required prior to qualification testing, when specified in the individual test paragraphs, to relieve stresses and establish uniformity. Reels or hubs of tape shall be rewound under a tension of 5 ounces ± 3 ounces. The reels or hubs of tape shall then be conditioned at 125° ± 5° F., 80 to 90 percent relative humidity (r.h.) for 3 hours. The tape shall then be conditioned at 75° ± 5° F., 50 ± 5 percent r.h. for a minimum of 3 hours. This temperature humidity cycle shall be repeated, with the transition times between temperature extremes not to exceed 60 minutes, until the tape has been subjected to a total conditioning time including transitions, of 20 hours. Upon removal, after the last cycle, the tape shall immediately be rewound a minimum of three times at a tension of 5 ounces ± 3 ounces. The tape shall then be held in a room conditioned at 75° ± 5° F., 50 ± 5 percent r.h. for a minimum of 24 hours before any tests are made. Preliminary conditioning shall not be required for quality conformance testing.

4.4.2 Test conditions. Unless otherwise specified in the contract or purchase order, all tests required by this specification shall be made at an ambient temperature of 75° ± 5° F., and a relative humidity between 45 and 55 percent, in a relatively dust free atmosphere.

4.4.3 Test samples. All physical tests required by this specification shall be performed on 1/4-inch tape samples.

4.4.4 Yield strength.

4.4.4.1 Apparatus. The apparatus for determining the specified tensile properties shall be as described in paragraph 3.2 of Fed. Test Method Std. No. 406, Method 1013.

4.4.4.2 Preparation of samples. Samples shall be conditioned as specified in paragraph 4.4.1.

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4.4.4.3 Procedure. A strip of tape not less than 7 inches in length shall be clamped in the grips set for an initial separation of 4 inches. The test specimen shall be clamped in the testing machine taking care to align the long axis of the specimen with an imaginary line joining the points of attachment of the grips to the machine. The grips shall be tightened evenly and firmly to the degree necessary to prevent slipping of the specimen during the test. The rate of grip separation shall be 2 inches per minute. Not less than five specimens of a particular type of tape shall be tested. A force strain diagram shall be obtained. Means shall be provided for calibration of the strain axis. The 1.0 percent offset yield strength shall be determined as described in ASTM Standard D638. The tangent (approximating the initial slope) to the force-strain curve shall be obtained by drawing a straight line between the origin and the point on the curve corresponding to one half the specified force (see table III). If the force-strain curve between these two points deviates by more than 0.1 percent elongation from the straight line drawn, a new straight line shall be drawn from the origin to one quarter of the specified force. The 1.0 percent offset yield strength determined in this manner and the force corresponding to 3 percent elongation shall equal or exceed the value listed in table III for the particular type of tape being evaluated. In the event that the breakage of a particular tape occurs prior to reaching the 1.0 percent offset yield strength or 3 percent elongation, the force at break shall equal or exceed the value listed in table III.

4.4.5 Elongation under stress.

4.4.5.1 Apparatus. The apparatus for this test shall consist of standard means of measuring length, positive nonslipping clamping devices, a 50 gram weight, and a weight as specified in 4.4.5.3.

4.4.5.2 Preparation of samples. Samples shall be prepared as specified in 4.4.1. Samples at least 24 inches long shall be clamped so as to hang in the test area (75° ± 5° F., 50 ± 5 percent r.h.) for at least 24 hours under no externally applied stress before tests are begun. Not less than five specimens of a particular type of tape shall be tested.

4.4.5.3 Procedure. Before any weight is hung on the samples, a mark shall be made approximately 20 inches from the point of clamping to measure the elongation and recovery of the sample. The distance between the mark and clamping point shall be measured accurately to the nearest 0.01 inch with an applied tension of 50 grams. This distance shall be taken as the base distance for calculation of residual elongation. When the measurement of the base distance has been made, the test shall begin. A test weight of 2-1/2 pounds shall be attached to the tape below the mark at zero time and allowed to hang undisturbed for 180 minutes ± 30 seconds, at which time the weight shall be removed from the tape. The tape shall be allowed to hang under its own weight for an additional 180 minutes ± 30 seconds. The distance between the mark and the point of clamping shall then be measured to the nearest 0.01 inch with an applied tension of 50 grams. The difference between the base distance and the final distance shall be expressed as a percent of the base distance to determine compliance with 3.3.2.

4.4.6 Humidity stability (cupping).

4.4.6.1 Apparatus. The apparatus for this test shall consist of the following: (If a controlled temperature and humidity room is available, this may be used in place of the chamber).

4.4.6.1.1 Chamber. The humidity chamber shall be of materials which are nonreactive to water vapor and potassium chloride solution and shall be constructed so that all joints are sealed tight when the chamber is closed. It shall consist of two separate compartments each approximately 11 by 13 by 7 inches. The front and back sides of the compartments shall be of a transparent material such as glass. The trays used to hold the chemicals inside the chamber shall measure approximately 10 by 10 by 4 inches and shall be made from aluminum, glass, or any nonreactive material. The trays shall be provided with removable perforated aluminum cover plates to permit placing tape holders above the conditioning chemicals with minimum interference with free air circulation. Means shall be provided for circulation of air within each compartment with a velocity of at least 20 feet per minute across both the conditioning chemicals and the tape holders.

4.4.6.1.2 Tape holders. Each tape holder shall, when filled, clamp the tape along its longitudinal axis and shall prevent any movement of the longitudinal axis of the tape by positively contacting it beginning at the end where the cupping is to be measured and continuing at least 1 inch along the tape length. The holder shall raise the tape at least 1/8 inch above the holder base plate and shall be separated from the adjacent tape holder by at least 5/8 inch, thus insuring that there will be no interference to cupping in either direction. For example, this may be accomplished by constructing the holder from two pieces of straight, rust or corrosion-resistant wire (the bottom piece being 1/8 inch in diameter and the top piece being 1/16 inch in diameter) 1-1/2 inches in length which are soldered or brazed together at the front tips, and then brazed onto a base plate of noncorrosive material. When the tape holders are in place in the chamber compartment and with tapes inserted for measurement, the longitudinal axis of the tapes shall be substantially horizontal.

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4.4.6.1.3 Measuring instrument and illumination. The measuring device shall be any optical system with a magnification of from 5 to 25 times, having at least one crosshair which can be referred to a clinometer or goniometer so as to measure the angle between the crosshair and a reference line, and having a focal length such that it can be focused on the near end of a tape when the tape is mounted in the chamber and the measuring device is placed in front of the chamber. A source of light shall be placed behind the chamber during measurement to outline the tapes when viewed through the measuring instrument.

4.4.6.2 Preparation of samples. No special preparation of the samples is needed, except as specified in 4.4.1. Not less than five specimens, per compartment, of a particular type of tape shall be tested.

4.4.6.3 Procedure. The test procedures shall be as follows:

(a) Prior to conditioning the samples, the initial tape cupping shall be measured on each test specimen. The measurement on each specimen is performed by viewing the ends of the specimen through the measuring instrument so as to measure the angle formed by the conjunction of lines constructed perpendicular to the edges of the viewed tape end. Local irregularities shall be averaged when setting the crosshair for these measurements.

(b) When using a two compartment chamber for conditioning the samples, one chamber shall be desiccant and the other chamber shall be humidifying. The desiccant chamber shall contain a desiccant anhydrous calcium chloride with indicating "Drierite" in the ratio of 3 to 1. The humidified chamber shall have a saturated solution (plus an excess) of potassium chloride in distilled water as the humidifying agent. The chambers containing the conditioning media shall be closed and the air circulated for at least 12 hours immediately preceding the insertion of the tapes to insure equilibrium. The test shall be run at a temperature of 90° ± 5° F. in both cabinets. The humidity conditions shall be 90 ± 5 percent in the wet chamber and 15 ± 5 percent in the dry chamber. If a two compartment chamber is used, a 6-inch length of tape shall be selected and cut into two 3-inch pieces with scissors (not a razor blade). These shall be mounted, as specified in 4.4.6.1.2, with the backing side down on two separate holders, one to be placed in the desiccant chamber and the other in the humidifying chamber, so that the measured ends are those made by the scissors cut in each case. The tape holders containing the tape specimens shall be placed in their respective sections. The compartments shall be closed, and the air shall be made to circulate for at least 16 hours before the measurements are made. (This shall apply to tests in a humidity room as well). At the end of conditioning, the differential cupping, the arithmetical difference in degrees between the angle measured on the desiccated tape and the angle measured in the same manner on the humidified tape, will be one of the criteria for acceptance or rejection of such tape. Since this test is designed to measure the effect of differential cupping, local irregularities shall be averaged when setting the crosshair for these measurements.

(c) Neither the "initial" cupping, "differential" cupping, nor the cupping under any one condition (wet or dry chambers) shall fail to meet the requirements specified in 3.3.3.

4.4.7 Fungus resistance. Resistance to fungus shall be determined in accordance with the method specified in MIL-I-631, except that a 2-inch diameter roll of tape shall be tested. The entire sample shall be tested in the rolled condition. Three specimens shall be tested for conformance to 3.3.4.

4.4.8 Layer-to-layer adhesion.

4.4.8.1 Apparatus. The apparatus for this test shall consist of a 1000 gram weight and a 1/2-inch diameter hollow tube. The tube shall be made of a nonoxidizing metal such as brass or corrosion resisting steel 1/2 inch in diameter and 4 inches in length, and shall weigh not less than 15 grams nor more than 30 grams. The tube shall be capable of being mounted in bearings so that it may be rotated freely around its central axis and easily removed from the bearings.

4.4.8.2 Preparation of samples. No special preparation of the sample is required, except as specified in 4.4.1. Not less than five specimens of a particular type of tape shall be tested.

4.4.8.3 Procedure. The test procedures shall be as follows:

(a) The 3-foot sample length of tape shall be fastened at one end, magnetic side down, to the 1/2-inch diameter hollow tube with a nonoozing adhesive material. The tube shall then be mounted in the bearings such that the tape shall hang free below the tube.

(b) Attached to the free end of the tape shall be the 1000 gram weight. A small strip of double-coated adhesive tape shall be affixed to the magnetic side of the tape one inch above the weight. The tube shall then be slowly and uniformly rotated so that the tape, held in tension by the weight, winds uniformly around the tube into a compact and even roll. The double-coated tape when wound into the test roll acts to secure the roll and prevent its unwinding when the weight is removed.

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(c) The tube supporting the rolled tape shall be removed from the winding setup and subjected to a heat and humidity cycle in which the first 16 to 18 hours shall be at 130° F. and 85 ± 5 percent r.h., while the final 4 hours shall be at 130° F. dry heat (less than 5 percent r.h.). During the humidification and dry heat cycle, provision shall be made so that the air surrounding the tube is constantly circulated to assure uniformity of conditions throughout the test area. At the end of the dry heat cycle, the rod shall be removed from the conditioned area and allowed to come to equilibrium with room conditions, approximaetly 75° F. at 50 percent r.h.

(d) To evaluate the tape for layer-to-layer adhesion, the end of the roll on the rod shall be carefully opened and the double-coated tab shall be removed. The rod shall then be held between the thumb and fingers and the untabbed tape shall be observed to note if the first two or three layers loosen up of their own accord; if this occurs, there is obviously no adhesion and the tape has passed the test. If no loosening or very little loosening of the outermost layers is observed, the free end of the tape shall be unwound slowly until 9 inches has been unwound. The free end shall be allowed to hang and the tape shall be observed to see if it will loosen by itself. If it will not unwind unaided, the rod, with the tape hanging freely, shall be slowly rotated in the direction of tape unwind. If the tape adheres to itself and refuses to begin to unwind after the rod has been rotated through one-fourth revolution or 90°, it shall be considered to have failed the test. After the rotation test has been made, the free end of the tape shall be held and the rod allowed to fall, thereby unwinding the tape. The unwound tape shall be checked for evidence of coating delamination and in this way the severity of adhesion is established. Any tape which will not self-unwind after rotating the rod through 90° or which shows any delamination except in the 2 inches nearest the rod shall be considered as having failed this test.

4.4.9 Longitudinal curvature.

4.4.9.1 Apparatus. No special apparatus, other than a 48-inch straightedge, is required.

4.4.9.2 Preparation of samples. No special preparation of the sample is required, except as specified in 4.4.1. Not less than five specimens of a particular type of tape shall be tested.

4.4.9.3 Procedure. Curvature shall be determined by constraining a sufficient length of tape to lie in a horizontal plane under zero tension and by measuring the maximum deviation of either tape edge from a 48-inch straightedge.

4.4.10 Wound pile envelope. The tape shall be examined for the wound pile envelope defects listed in table VII to determine compliance with 3.3.7.

4.4.11 Anchorage. A visual examination shall be made of the tape after completion of all tests to determine compliance with 3.3.8.

4.4.12 Splices. The tape shall be examined to determine compliance with 3.3.9.

4.4.13 Workmanship and general examination. The tape, as initially received from the manufacturer, shall be visually examined to determine that the reels or hubs and the tape are free from the applicable defects listed in table VII.

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TABLE VII. General examination defects

Examine	Defect
Wind	Oxide surface not toward the hub.
Wound pile envelope	Tape pile not smoothly wound; loose, with visible folds, buckling, cinching, spoking, or gaps between the tape layers. Protruding tape edges, tape layers, or groups of layers. Plane of the tape pile not perpendicular to the reel axis. Tape not wound on reel that meets the requirements of 3.1.4.4.
Appearance	Tape surfaces not clean; presence of dirt, dust, lint, fuzz, or other foreign matter. Presence of blemishes, holes, tears, creases, or wrinkles, split or ragged edges.
Construction	Tape does not unroll evenly and uniformly; tape sticks; tape affected by commonly used head cleaners.
Condition, as received	Improper packaging. Improper marking. Damaged contents.

4.5 Magnetic tests and measurements.

4.5.1 Definitions.

4.5.1.1 Primary reference tape. The primary reference tape is a special unrecorded length of tape used as a reference to establish operating bias current, maximum recording level, and standard output level for the reference recorder used in the tests described herein. This reference is maintained at the General Services Administration, Magnetic Surfaces Laboratory.

4.5.1.2 Secondary reference tape. The secondary reference tape is an unrecorded length of tape, the magnetic characteristics of which have been calibrated against those of the primary reference tape. The secondary reference tape is used for preliminary adjustments of the reference recorder and for certain magnetic tests hereinafter specified. Secondary reference tapes are available from the Director, Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C. 20406. The qualifying activity reserves the right to levy a charge to recover the cost of calibrating secondary reference tapes.

4.5.1.3 Operating bias current. Operating bias current is that bias current through the recording head which will give peak output from the primary reference tape at maximum recording level. Operating bias current is set using a secondary reference tape (see 4.5.1.2). The amount of bias current for the secondary reference tape that corresponds to the operating bias current for the primary reference tape is given on the calibration sheet which accompanies the secondary reference tape.

4.5.1.4 Maximum recording level. The maximum recording level is that input level which in conjunction with operating bias current will produce a permanent induction in the primary reference tape such that on playback of a 1000 Hz signal, the output signal will have 3 percent third harmonic distortion. Maximum recording level is set in conjunction with operating bias current using the secondary reference tape (see 4.5.1.2). The amount of harmonic distortion for the secondary reference tape that corresponds to 3 percent third harmonic distortion for the primary reference tape is given on the calibration sheet which accompanies the secondary reference tape.

4.5.1.5 Standard recording level. The standard recording level is 15 db below the maximum recording level.

4.5.1.6 Maximum output level. The maximum output level is that output voltage measured on playback of a 1000 Hz signal recorded on the primary reference tape at maximum recording level with operating bias current.

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4.5.1.7 Standard output level. The standard output level is that output voltage measured on playback of a 1000 Hz signal recorded on the primary reference tape at standard recording level with operating bias current. Standard output level is determined using the secondary reference tape (see 4.5.1.2). The output level for the secondary reference tape that corresponds to the standard output level for the primary reference tape is given on the calibration sheet which accompanies the secondary reference tape.

4.5.2 Reference test recorder. For the test specified in 4.5.4, a reference magnetic tape recorder shall be used. The reference test recorder shall have the following minimum features:

(a) The recorder shall provide a tape speed of 7.5 inches per second ± 0.1 percent.

(b) The frequency response shall be flat to within ± 2 db over the range 30 Hz to 15 kHz at a tape speed of 7.5 inches per second using the primary reference tape (see 4.5.1.1).

(c) The harmonic distortion introduced by amplifiers and heads shall not exceed 0.5 percent at all levels and frequencies used in the magnetic tests.

(d) The overall signal-to-noise ratio shall be 60 db or higher. The overall signal-to-noise ratio is defined as the ratio of a maximum output level to the total unweighted playback noise after erasure of a signal recorded at maximum recording level and in the absence of a new signal. Thus, bias and erase noises are included as well as the playback and record amplifier noises. All frequencies between 30 Hz and 20 kHz shall be measured.

(e) The flutter shall be less than 0.15 percent r.m.s. at a tape speed of 7.5 inches per second, measuring all components from 0.5 Hz to 300 Hz.

(f) The recorded track width shall be the full width of tape.

(g) The erase signal frequency shall be 95 kHz or higher. The signal strength shall be such as to reduce the level of a signal recorded at maximum recording level on the primary reference tape by at least 65 db.

(h) The bias frequency shall be 95 kHz or higher, and adjustable in amplitude.

(i) The pre-equalization characteristic of the reference test recorder shall be as shown in figure 1.

(j) The post equalization characteristic of the reference test recorder shall be within the limits shown in figure 2, and such that the overall response is in accordance with 4.5.2.

(k) There shall be a separate head for each of the functions: erase; record; and playback (3M part no. 84-5950-1227 or equivalent); and, the recorder shall be configured so that playback may be accomplished simultaneously with record.

(l) The recorder shall accommodate all reels and hubs specified in 3.1.4.4 up to 10.5-inches in diameter.

(m) A professional type commercial recorder (such as 3M series 400 or equivalent) which conforms to the requirements specified herein may be used for the applicable tests specified in 4.5.4.

4.5.3 Test setup. The test setup for the reference test recorder prior to performing the magnetic tests shall be as follows:

(a) All magnetic tests shall be conducted at a tape speed of 7.5 inches per second ± 0.1 percent and with a 0.250 inch track width.

(b) The recorder shall be thoroughly cleaned and demagnetized and the record and playback heads shall be adjusted for correct azimuthal orientation of their respective gaps.

(c) The tension of the recorder shall be measured to determine that the recorder is in proper adjustment.

(d) The maximum recording level and operating bias current shall be established for the reference test recorder using the secondary reference tape (see 4.5.1.2) and corrected to the primary reference tape.

(e) For all tests, the reference test recorder shall be terminated with its proper load impedance.

(f) All test signals shall be sine wave signals.

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(g) Before tests, the recorder shall have had the equivalent of at least 20,000 feet of tape passing the heads under record or reproduce conditions.

(h) For the magnetic measurements, the tape samples shall not require any preliminary conditioning.

4.5.4 Test procedures.

4.5.4.1 Sensitivity. The standard output level (see 4.5.1.7), shall be established to calibrate the reference test recorder for the sensitivity measurement. A 1000 Hz signal shall then be recorded on the tape to be tested at the standard recording level and at operating bias current. This tape shall be played back and the output voltage of the playback amplifier of the recorder measured in db. Correction shall be made for the response of the reference recorder with a secondary reference tape at operating bias current. The third harmonic distortion shall also be measured.

4.5.4.2 Frequency response. The procedure specified in 4.5.4.1 shall be repeated at all frequencies specified in table IV. The output of each frequency shall be normalized to the 1000 Hz output and expressed in db. Correction shall be made for the response of the reference test recorder with a secondary reference tape at operating bias current.

4.5.4.3 Output level uniformity. The frequencies specified in table IV shall be recorded at standard recording level and with operating bias current. On playback, the output voltage variations shall be measured by means of a linear recording oscillograph having a response which is uniform to at least 200 Hz. The tape uniformity, over a tape length of at least 50 feet, shall be measured by determining the peak variation in output as a percentage of maximum peak output. The tape shall meet the requirements specified in table IV over any 10 second interval, with all non-recurring pips being disregarded.

4.5.4.4 Signal to direct current (dc) noise ratio. A 1000 Hz signal shall be recorded at the maximum recording level with operating bias current. The tape shall be played back and the voltage output of the playback amplifier of the reference test recorder measured in db using a 250 Hz high-pass filter which provides no more than a 3 db attenuation at the cutoff frequency and has an 18 db per octave rolloff characteristic. This value shall be taken as the signal level. Third harmonic distortion shall also be measured. The tape shall then be externally erased, and a direct current supplied to the record head equal to the r.m.s. value of the audio current at maximum recording level. Under these conditions the tape shall be recorded on the reference test recorder with operating bias current. The output voltage on playback, measured as specified herein, shall be taken as the dc noise level. The signal to dc noise ratio is the value in db of the signal level minus the value of the dc noise level.

4.5.4.5 Layer-to-layer signal transfer. The tape shall be externally erased. With the reference test recorder operating at maximum recording level with operating bias current, and with erase head operating, ten layers of tape shall be recorded with no signal input to the reference test recorder and shall be wound on a reel or hub classified as ER-250-PC or EH-250-RC in accordance with W-R-175. The record level shall be increased to maximum recording level at operating bias current with a 1000 Hz signal, and one additional layer recorded and wound on the reel or hub. The record level shall be returned to zero and ten additional layers shall be recorded and wound on the reel or hub. The recorded tape shall be conditioned at a temperature of 150° F. for a period of 4 hours. The tape shall then be played back and the output of the playback amplifier shall be connected to a 1000 Hz band-pass filter having a 100 Hz passband. The level of the recorded signal and the maximum level of the signal resulting from signal transfer shall be measured at the output of the filter.

4.5.4.6 Ease of erasure. The tape shall be externally erased. A frequency of 1000 Hz shall be recorded at maximum recording level with operating bias current and with erase head operating. The tape shall then be conditioned at a temperature of 150° F. for 4 hours. The tape shall be played back and the output of the playback amplifier shall be connected to a 1000 Hz band-pass filter having a 100 Hz passband. The output of the filter shall be measured. The tape shall then be passed through a solenoidal coil producing a 60 Hz ac field of 1000 oersteds (peak value) after which the level of the residual signal on playback shall be measured through the 1000 Hz band-pass filter. The difference between the recorded signal level and the residual signal level is the effective reduction in signal or ease of erasure.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.1).

5.1.1 Level A. Packaging shall be in accordance with the level A packaging requirements of MIL-E-17555.

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5.1.2 Level B. Packaging shall be in accordance with the level A packaging requirements of MIL-E-17555 except that the bag shall conform to L-P-378, shall be a minimum of 2 mils thick and may be closed with a wire or clip of the type normally used for closing this type of bag.

5.1.3 Level C. Packaging shall be in accordance with the manufacturer's standard practice providing that this will insure protection for the tape during shipment and provides for safe delivery to its destination.

5.2 Packing. Tape packaged in accordance with 5.1 shall be packed in accordance with the level A, B or C requirements of MIL-E-17555 as specified (see 6.1).

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2 as specified (see 6.1).

5.3.1 Civil agencies. In addition to any special markings required by the contract or order, all markings shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special markings required by the contract or order, all marking shall be in accordance with MIL-STD-129.

5.3.3 Special marking. In addition to the marking required by 5.3, each intermediate and shipping container shall be marked with the following information:

(a) Tape, recording, audio, magnetic oxide-coated.

(b) Classification.

(c) Name of manufacturer.

(d) Manufacturer's identifying tape name/number.

(e) Federal Stock No. (if available).

6. NOTES

6.1 Ordering data. Purchasers should exercise the preferred options permitted herein, and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Quantity of tape required.
(c) Tape designator (see 1.2.1, 1.2.2, and 1.2.3).
(d) Reel or hub required (see 1.2.5, 1.2.6, and 3.1.4.4).
(e) Length required (see 1.2.7 and 3.2.1).
(f) Inspection responsibility (see 4.1).
(g) Packaging and packing required (see 5.1).
(h) Special markings, if required (see 5.2).

6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bid, been tested and approved for inclusion in Qualified Products List QPL-W-T-001572, whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Federal Supply Service, General Services Administration, Washington, D. C. 20406, and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Application by Manufacturers for Inclusion on Federal Qualified Products Lists (QPL's)", (see 6.3).

6.3 Copies of "Provisions Governing Application by Manufacturers for Inclusion on Federal Qualified Products Lists (QPL's)" may be obtained upon application to Director, Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C 20406.

6.4 Supersession data. This specification supersedes those portions of W-T-0070 (NAVY-ships) covering tapes, audio type (W-T-0070/1 and W-T-0070/2).

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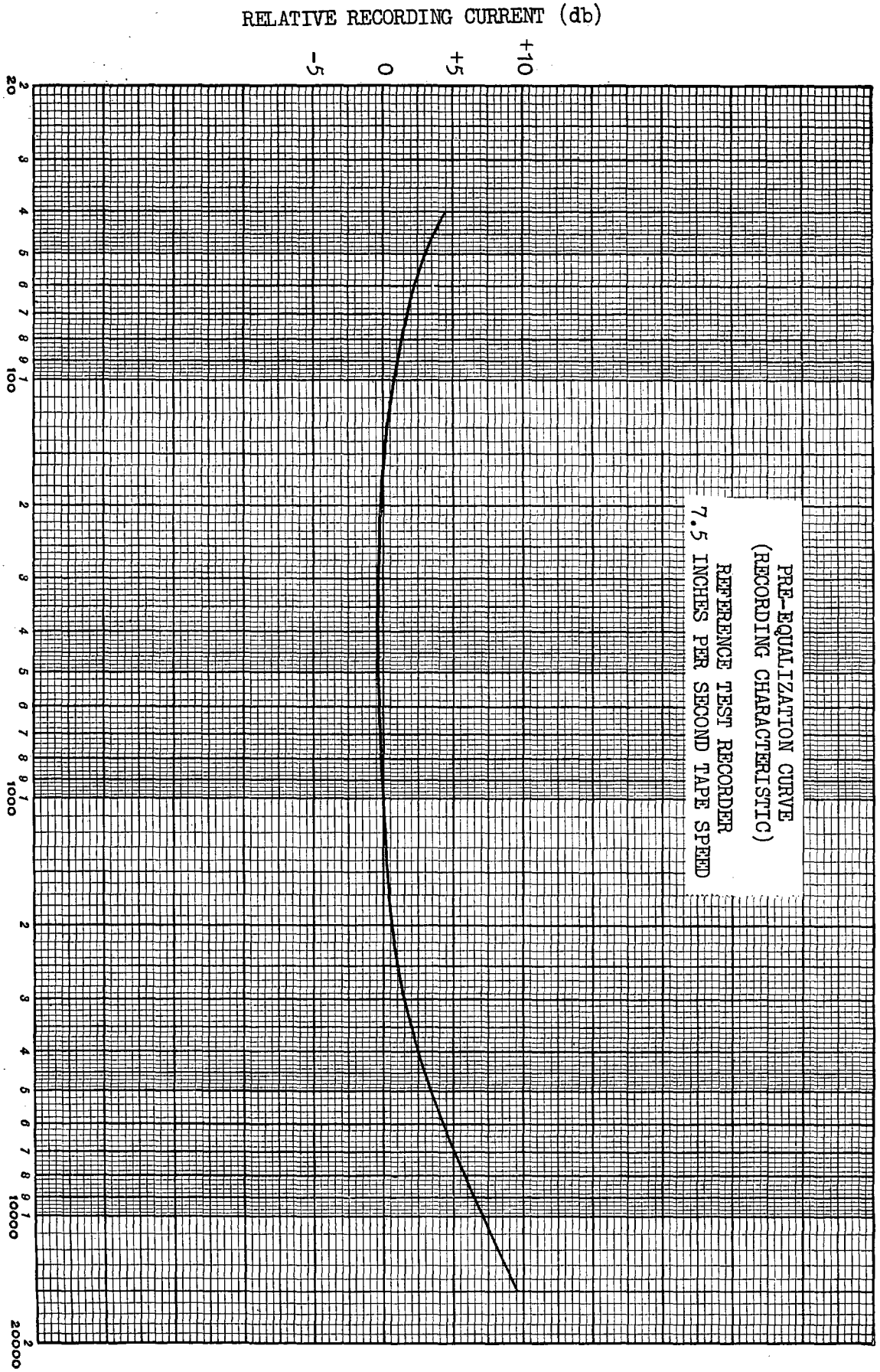


FIGURE 1 - PRE-EQUALIZATION CURVE (RECORDING CHARACTERISTIC)

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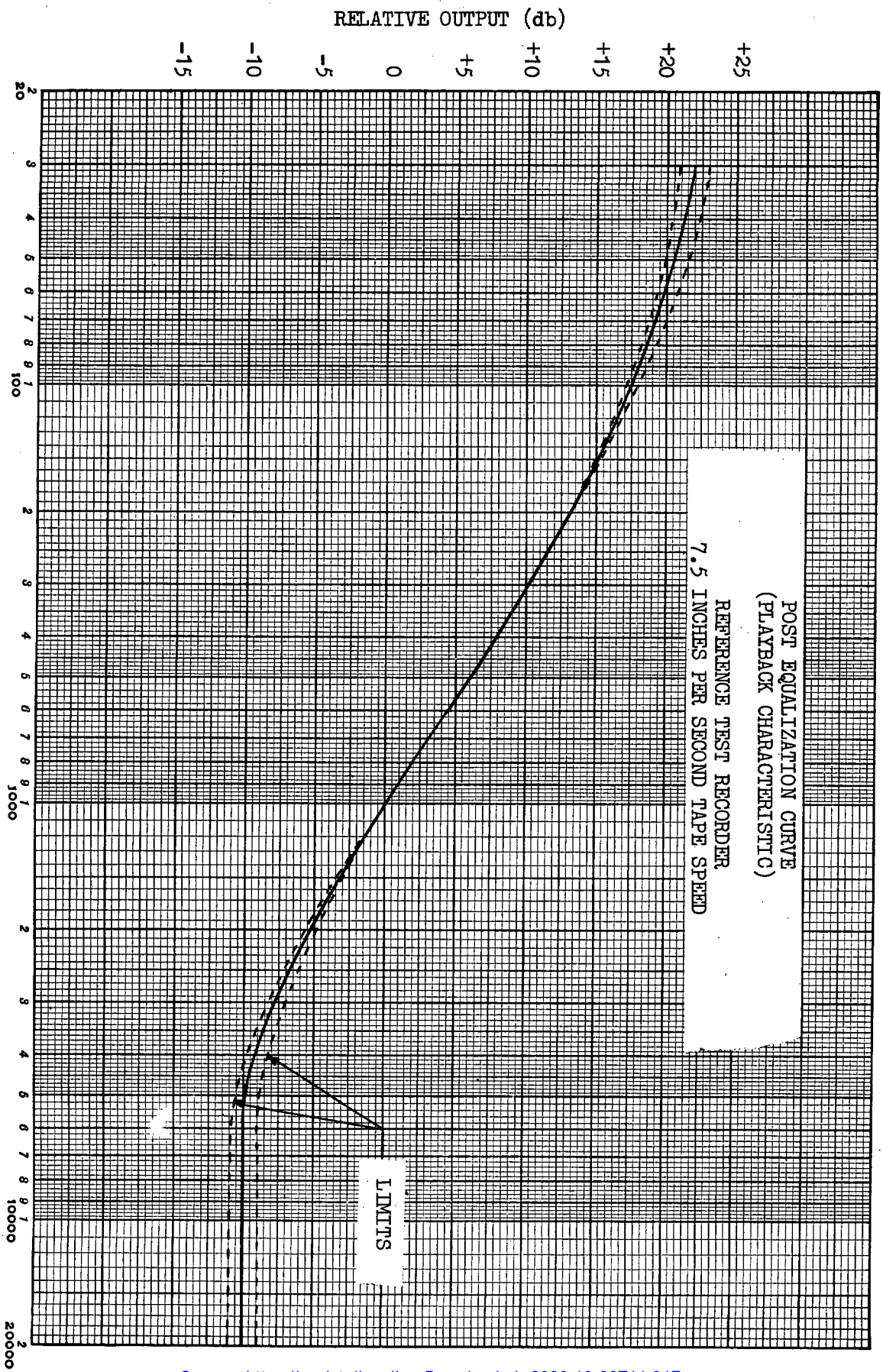


FIGURE 2 - POST EQUALIZATION CURVE (PLAYBACK CHARACTERISTIC)

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